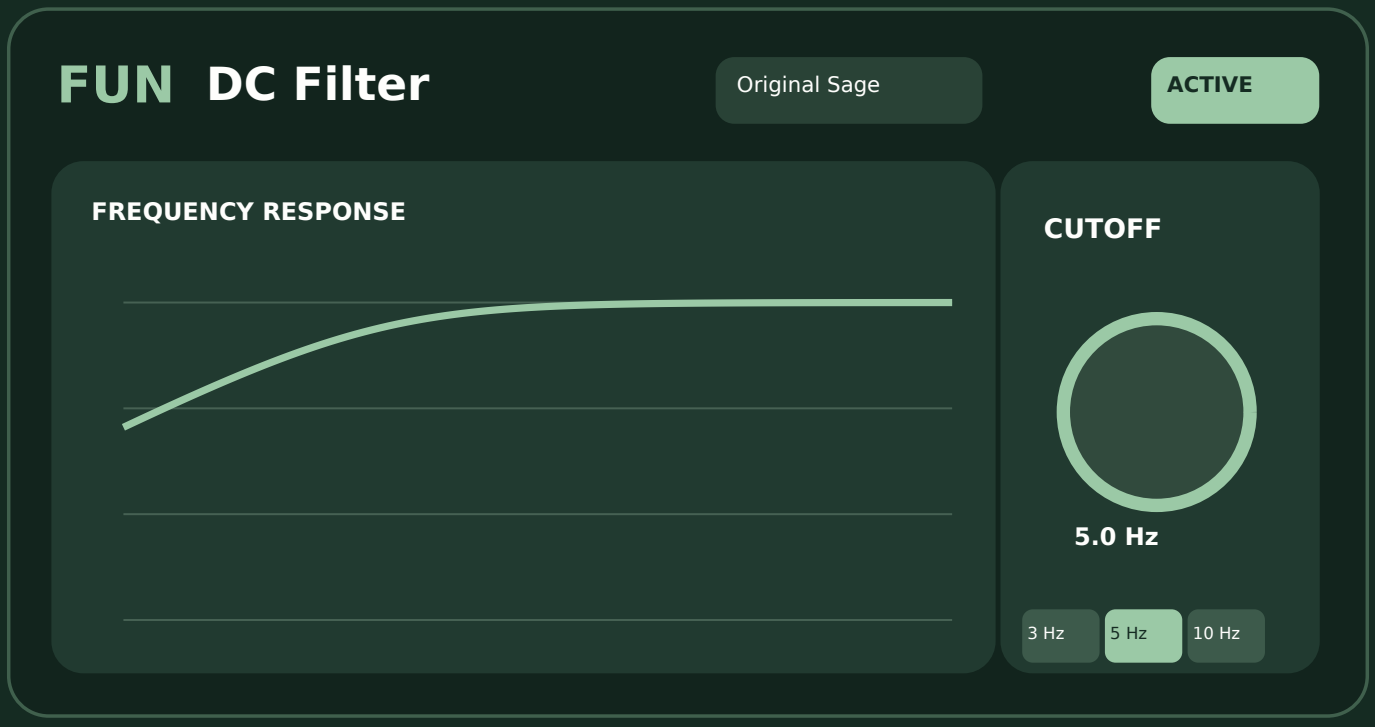
FUNtasia Audio

FUN DC Filter

User Manual

A focused DC and subsonic cleanup tool for music production.



Controls · Signal processing · DAW workflow

FUN DC Filter is a first-order high-pass filter designed to reduce DC offset and unwanted subsonic content. Choose only the processing you need, and monitor its effect on your low end.

For v1.0.3 RC2

01 Getting started

Start with three simple steps.

1 **Insert**

Load FUN DC Filter on an audio track, bus, or master channel in your DAW.

2 **Set Cutoff**

Start at the 5.0 Hz default. Use the 3 / 5 / 10 Hz shortcuts or the CUTOFF knob as needed.

3 **Compare**

Toggle ACTIVE to hear the difference. ON enables the filter; OFF passes the dry signal.

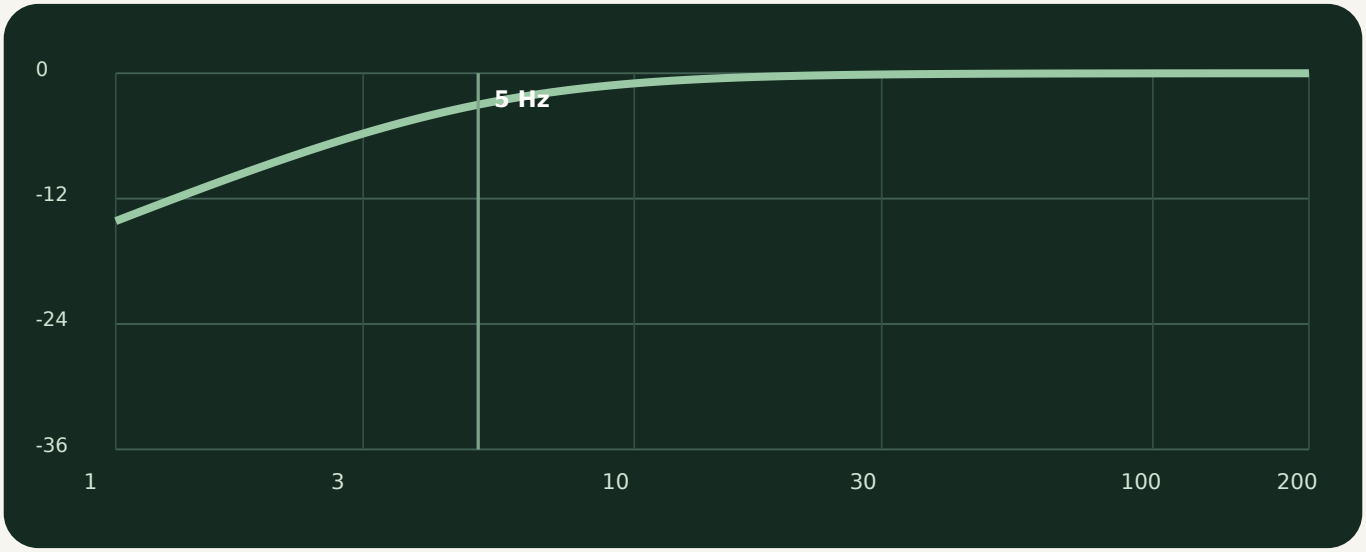
Controls at a glance

CUTOFF	1.0-100.0 Hz in 0.1 Hz steps; 5.0 Hz default.
3 / 5 / 10 Hz	One-click frequency shortcuts.
Value field	Double-click the displayed number for direct entry.
ACTIVE	Enables/disables the filter with a roughly 15 ms transition.
Theme selector	Click the color name at the top to choose among seven themes.
Resize corner	Drag the lower-right corner; the 2:1 aspect ratio is fixed.

The plugin interface uses English labels in every installation. This manual keeps the on-screen names unchanged.

02 Reading the response

FREQUENCY RESPONSE plots the calculated response for the current Cutoff and sample rate. It is not a live analyzer of the incoming audio.



X axis	Logarithmic frequency scale: 1-200 Hz.
Y axis	Gain: 0 to -36 dB. Values below the range are clipped visually.
Line and marker	The colored line is the calculated response; the vertical marker identifies the Cutoff.

Inside the processing path

- First-order TPT high-pass filtering, with approximately 6 dB/oct attenuation. Being minimum-phase, it also changes phase near the cutoff.
- Cutoff changes are smoothed for about 15 ms; ACTIVE transitions use a separate roughly 15 ms crossfade.
- With ACTIVE OFF, the filter state still tracks the signal while ordinary finite audio passes through dry.
- Mono and stereo are supported. Left and right filter states are independent in stereo.

03 Working in a DAW

Choose the Cutoff for the material, then confirm the result by listening and metering.

01

Reduce DC offset

Try 3-5 Hz first. Observe any waveform-center bias or unnecessary subsonic energy.

02

Kick and bass

Protect the lowest useful fundamentals. Above 10 Hz, listen carefully for changes in impact and weight.

03

A/B listening

Switch ACTIVE on and off to compare level and low-frequency content. The switch is smoothed, but the filter still has a phase response.

04

Automation and recall

CUTOFF and ACTIVE support DAW automation. Cutoff, ACTIVE and the GUI theme are saved in plugin state.

Production notes

- A low Cutoff does not guarantee zero audible change; choose a setting that preserves the low end you want.
- When automating large Cutoff moves, audition transitions in the actual project.
- Each plugin instance maintains independent settings and its own GUI theme.

04 Specifications & help

Format	VST3 (Windows build); Standalone target in the project
Channel layouts	Mono in/out; stereo in/out
Audio precision	32-bit float and 64-bit double processing
Filter topology	First-order TPT high-pass, ~6 dB/oct
CUTOFF	1.0-100.0 Hz; 0.1 Hz steps; default 5.0 Hz
Transitions	Cutoff smoothing 15 ms; ACTIVE crossfade 15 ms
Host-reported	Latency 0 samples; tail 0
GUI dimensions	Default 616 x 308 px; 560 x 280 to 1680 x 840 px
Themes	Original Sage, Charcoal, Deep Brown, Lapis Lazuli, Lavender, Coral Red, Mikan Orange
Saved state	Cutoff, ACTIVE and GUI theme

Troubleshooting

No audible difference

Check that ACTIVE is ON. Signals with little DC offset or subsonic energy may change very little.

Unexpected window size

Some DAWs restore the last editor size. Use the lower-right resize corner.

Opening old projects

Duplicate the DAW project before testing. Legacy states without a saved theme open in Original Sage.

Based on the v1.0.3 RC2 source. The graph shows a calculated response; this guide does not certify subjective sound, external measurements or every host configuration.